# Root Cause Analysis Report

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|---|---|
| Part | MS-4505-001 Sheet Metal Bracket |
| Material / process | SPCC Steel / bending |
| Case | rca_5bccb3b439cd450f9788a9999e59597b |
| Date | 2026-11-05 06:24:58 |
| Status | AI proposal, pending expert review |

**WARNING:** the AI model was not used for this analysis (fake-offline). The text below is a generic offline template, not a result.

## 1. Problem

| Defect | Severity | Confidence | Detail |
|---|---|---|---|
| crack | critical | 93% | |
| crack | critical | 91% | |
| crack | critical | 85% | |
| crack | critical | 82% | |
| crack | critical | 60% | |
| crack | critical | 29% | |
| crack | critical | 15% | |
| crack | critical | 13% | |
| dent | major | 12% | |
| crack | critical | 11% | |

## 2. Root cause

**Cause:** Improper Bending Force / Over Bending

**Reason:** The detected deformation and dimensional inaccuracy strongly suggest excessive or improperly calibrated bending force. Retrieved RCA documents confirm this pattern for SPCC steel bending operations.

**Effect:** Deformation, Dimensional Inaccuracy

**Confidence:** 66%

**Supporting evidence**
- **seed-001**: Improper Bending Force or Over Bending causes deformation and dimensional inaccuracy. Precautions: optimize bending parameters, use proper tooling, calibrate machines regularly.
- **seed-004**: Springback in sheet metal bending causes dimensional inaccuracy. Overbend compensation and material property testing are required.

## 3. Solution

### *Corrective actions (fix this occurrence)*

1. Re-set the press-brake bending force and angle to the material specification
2. Re-inspect the affected lot for deformation and dimensional deviation

### *Preventive actions (stop it recurring)*

1. Add bending-force verification to the machine calibration schedule
2. Control bending parameters per material batch

### *Precautions*

- Optimize bending parameters to material specification

- Use proper tooling and punch/die sets
- Regular machine calibration and force verification

### *How to verify the fix*

Re-inspect the next 30 parts; no deformation and all dimensions within tolerance

## 4. Other causes considered

| Rank | Cause | Confidence | Reason |
|---|---|---|---|
| 2 | Material Inconsistency / Hardness Variation | 63% | Hardness variation in the incoming material can cause unpredictable springback and cracking. Material certificate verification is required. |
| 3 | Tool Wear / Die Damage | 60% | Progressive die wear causes scratches, dents, and edge burrs. Preventive maintenance schedule should be implemented. |

## 5. Traceability

AI model: fake-offline. Expert edited the cause: no.
*This analysis was drafted with AI assistance from the retrieved knowledge-base evidence. It must be reviewed and confirmed by a qualified quality expert before corrective action is taken.*